

eRoyal Housing Society Management System

A role-based digital platform for modern society operations.

eRoyal centralizes residents, bills, complaints, announcements, marketplace listings, vehicle registration, gate entry logs, image uploads, and AI-assisted support in one secure Firebase-powered application.

1 One Platform, Three Protected Portals

Admin Portal

- Create and manage society users
- Generate monthly resident bills
- Verify uploaded payment proofs
- Resolve complaints and monitor logs

Resident App

- View announcements and bills
- Upload payment proof images
- Submit complaints with photos
- Post marketplace and vehicle details

Security Gate

- Record entry and exit activity
- Track active vehicles at gate
- Handle residents, visitors, services
- Use OCR for license plate scanning

Login

> Role Detection

> Protected Portal

> Firebase Sync

> Live Records

2 Core Project Modules

U User And Roles

Admin manages residents, admins, and security staff with authenticated role-based access.

B Billing

Monthly bills are generated, sent to residents, and verified using uploaded payment proof.

C Complaints

Residents submit complaints with photos, while admin tracks status and resolution notes.

A Announcements

Society notices are published digitally so residents can receive updates from one place.

M Marketplace

Residents can create property sale or rent listings with descriptions and photos.

V Vehicle Gate

Security records resident, visitor, and service vehicle entries with live active status.

AI Assistant + OCR

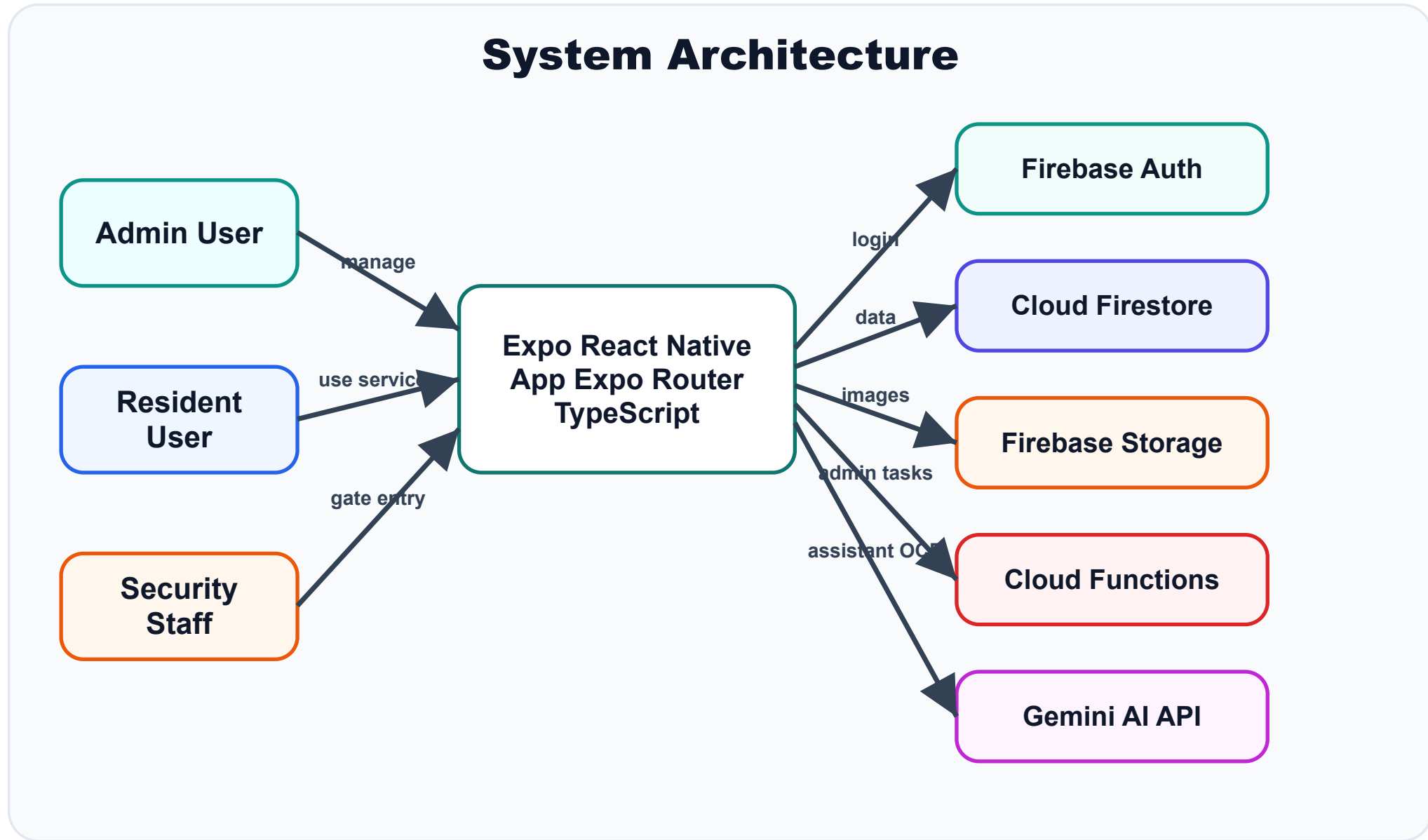
Gemini supports resident assistance and camera-based license plate recognition.

F Firebase Cloud

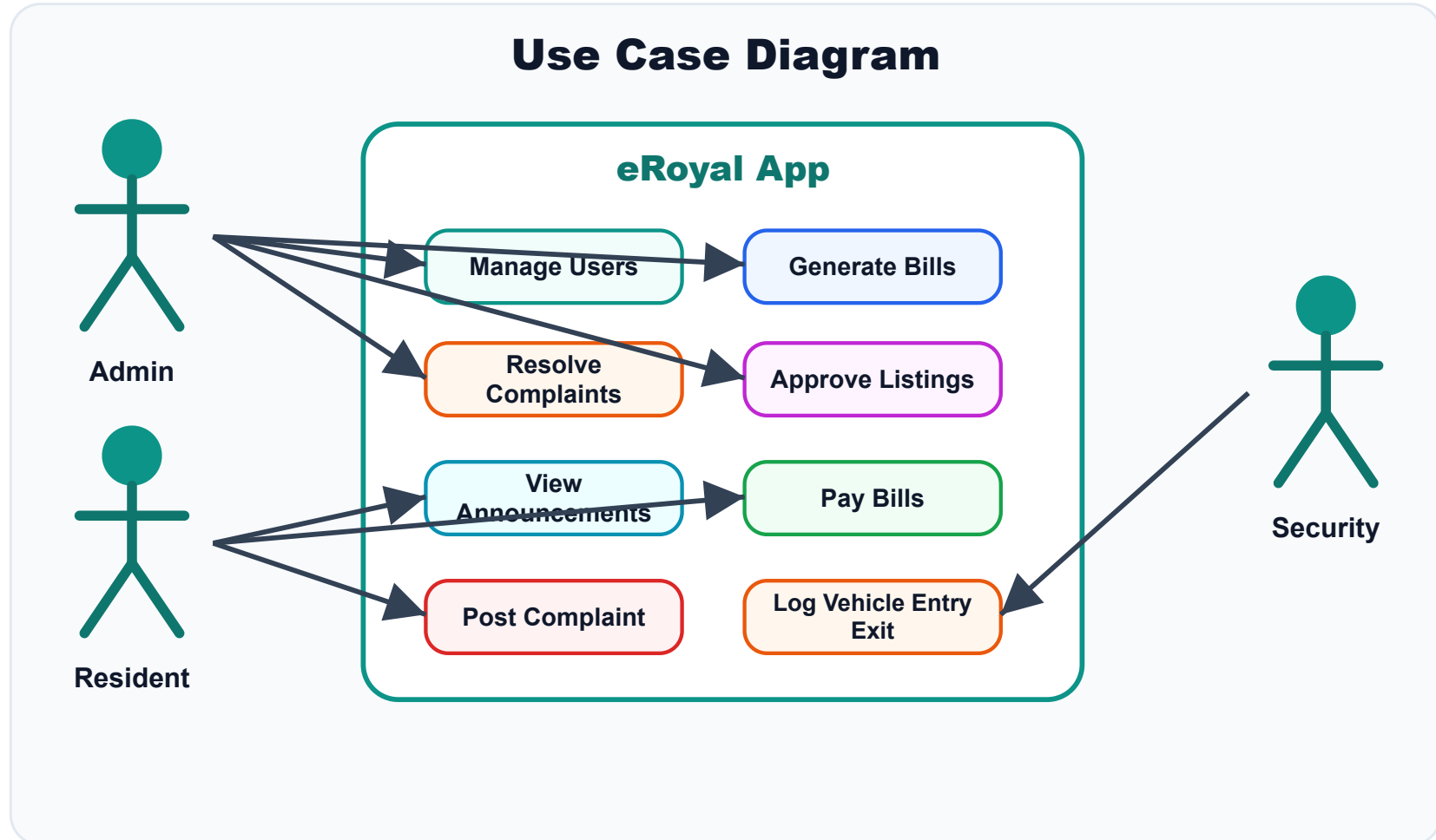
Authentication, Firestore, Storage, security rules, and real-time data updates power the app.

3 System Architecture And Use Case View

Architecture: React Native + Expo + Firebase



Use Cases: Admin, Resident, And Security Workflows



React Native

Expo Router

TypeScript

Firebase Auth

Cloud Firestore

Firebase Storage

Gemini API

Role Security

4 Project Impact

Less Paperwork

Manual registers and scattered records are replaced by one digital workflow.

More Transparency

Residents can track bills, complaints, vehicles, and notices from the app.

Faster Operations

Admin and security staff get live records, role-based dashboards, and image support.

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